

Last Time

Effect Sizes

Three-Way ANOVA in R

Repeated Measures ANOVA

- Partitioning the Variance

- Post-hoc analysis

- Effect Size

A Few Words About Regression

A Few Words About Bayes

- ▶ Bayes is the future!
- ▶ JASP is a statistics package that can do most standard analyses in a Bayesian fashion, and it's free!
- ▶ PSY 6820 is the Introductory Bayes class. ~~Jay Myung teaches a Bayesian modeling class focusing on models of cognition.~~

What is Bayes?

All statistical procedures depend on a model of the data.

- ▶ A model tells us what the distribution of the data is (Binomial, Poisson, Normal, etc.).
- ▶ Distributions have parameters and the goal of inference is to figure out what values those parameters have and how they change over experimental conditions (e.g., $H_0 : \mu_1 = \dots = \mu_J$).
- ▶ Whereas standard inferential procedures assume these parameters to be fixed constants, Bayesian procedures assume that they are random in the same way the data are.

Three Components

- ▶ Our prior beliefs about the parameters θ : $\pi(\theta)$ is the *prior distribution*.
- ▶ Our model of the data: $f(X | \theta)$, which is rewritten as the *likelihood* of the data for different values of θ , or $\mathcal{L}(\theta | X)$.
- ▶ The updated or *posterior* beliefs about θ having observed the data X : $\pi(\theta | X)$ is the *posterior* distribution.

Note that the *frequentist* tradition assumes that all error is due to sampling variability and all procedures are based on what will happen if we sample from the data distribution a very large number of times. For Bayesian procedures the data are fixed.

Bayes Theorem

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B \mid A)P(A) + P(B \mid \bar{A})P(\bar{A})}$$

$$\pi(\theta \mid X) = \frac{\mathcal{L}(\theta \mid X)\pi(\theta)}{\int_{\theta} \mathcal{L}(\theta \mid X)\pi(\theta)d\theta}$$

$$\pi(\theta \mid X) \propto \mathcal{L}(\theta \mid X)\pi(\theta)$$

How Does This Work?

You administer a 20-item true false exam to your class. A student approaches you after you return the grades to complain. “I studied so hard, but I failed this exam with a score of 11/20 or 55%.” Do you have reason to believe that the student actually studied?

Let $\theta = Pr(\text{getting a question correct})$, so

$$H_0 : \theta \leq 0.50 \text{ and } H_1 : \theta > 0.50$$

are the hypotheses that describe only guessing ($\theta = 0.5$) or knowing at least a little about the material ($\theta > 0.5$).

The Prior and the Likelihood

The model of the data is the Binomial distribution, where $N = 20$ (the number of questions) and θ is the probability of a “success,” a correct response.

Suppose you don't want to make any strong assumptions about θ . Even though this person has come to class infrequently, your prior beliefs about θ will permit you to consider any value between 0 and 1 (the Uniform[0,1] distribution).

$$X = 11$$

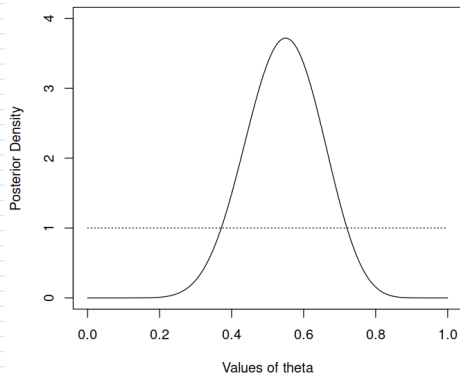
$$N = 20$$

$$\pi(\theta) = 1 \text{ for } \theta \in [0, 1]$$

$$\mathcal{L}(\theta \mid X = 11) = \binom{20}{11} \theta^{11} (1 - \theta)^9$$

$$\pi(\theta \mid X = 11) \propto \theta^{11} (1 - \theta)^9 \rightarrow \theta \mid X = 11 \sim \text{Beta}(12, 10)$$

Conclude What?



$$Pr(H_0 \mid X = 11) = 0.33$$

$$Pr(H_1 \mid X = 11) = 0.67$$

Bayes Factor

The *Bayes Factor* quantifies the support of one model (H_0) vs. a competing model (H_1) in light of the evidence provided by the data X .

$$B_{10} = \frac{Pr(X|H_1)}{Pr(X|H_0)} = \frac{Pr(H_1|X) Pr(H_0)}{Pr(H_0|X) Pr(H_1)}$$

$$Pr(H_1) = Pr(H_0) = 0.5 \text{ (Prior)}$$

$$Pr(H_1|X) = 0.33$$

$$Pr(H_0|X) = 0.67$$

$$\rightarrow B_{10} = 0.67/0.33 \approx 2$$

	Column 1
1	1
2	1
3	1
4	1
5	1
6	1
7	1
8	1
9	1
10	1
11	1
12	0
13	0
14	0
15	0
16	0
17	0
18	0
19	0
20	0

Bayesian Binomial Test



☐ Show all options

```
jaspFrequencies::BinomialTestBayesian(
  data = NULL,
  version = "0.95",
  formula = ~ "Column 1",
  alternative = "greater",
  priorPosteriorPlot = TRUE)
```

Ctrl + Enter to apply

Test value: 0.5

Alt. Hypothesis

Direction

- ☐ > Test value
- ☒ > Test value
- ☐ < Test value

Prior

Beta prior: parameter a 1

Beta prior: parameter b 1

Bayes Factor

- ☒ BF_{±0}
- ☐ BF_{±1}
- ☐ Log(BF_{±0})

Plots

- ☒ Prior and posterior
- ☒ Additional info
- ☐ Sequential analysis
- ☐ Descriptive plots

Credible interval 95 %

Bayesian Binomial Test

Bayesian Binomial Test

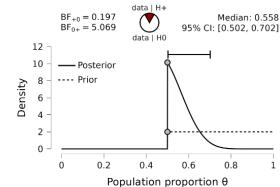
	Level	Counts	Total	Proportion	BF _{±0}
Column 1	0	9	20	0.450	0.197
	1	11	20	0.550	0.397

Note: For all tests, the alternative hypothesis specifies that the proportion is greater than 0.5. The shape of the prior distribution under the alternative hypothesis is specified by Beta(1, 1).

Inferential Plots

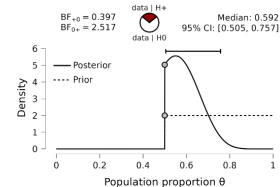
Column 1 - 0

Prior and Posterior



Column 1 - 1

Prior and Posterior



Thank You!

